

RESEARCH ARTICLE

Sample Complexity of Calibration versus Model-Free Learning in Cartea–Jaimungal–Penalva Optimal Liquidation

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ABSTRACT

We construct a sample-complexity benchmark for optimal liquidation in the [Cartea et al. \(2015\)](#) (CJP) Poisson-fill limit-order/market-order framework, comparing model-free reinforcement learning (RL) against a parametric model-based calibrator that estimates (κ, λ) and plans by one finite-difference solve of the CJP quasi-variational inequality. In the CJP Chapter 8 scale ($Q_0 = 10$, $T = 60$ s), the plug-in MLE attains about +0.194 dollars per episode relative to TWAP—matching the FD-optimal premium to within Monte-Carlo error. By contrast, Double DQN saturates near +0.129 and a structure-aware hybrid PPO reaches about +0.135 after removing a depth-floor artefact, but clears inventory in only about 23% of evaluation paths. We then add an institutional-scale stress test ($Q_0 = 100$, $T = 600$ s). In this larger regime, cash premium alone is statistically inconclusive: Hybrid PPO has the largest point estimate, but no learned-agent difference from Plug-in MLE survives Holm-corrected pairwise testing, and Plug-in MLE remains the only method combining FD-level value accuracy, high pre-terminal clearance, and negligible compute. We attribute these findings to three forces: (i) a reward-shaping identity (Theorem 5.1) under which the expected return of any policy equals the reduced value function $h(t, q)$; (ii) a CIR-type stochastic intensity extension validated at $\lambda_0 \in \{0.7\lambda, \lambda, 1.3\lambda\}$ for $\sigma_\lambda \in [0, 0.1]$; and (iii) a parameter robustness analysis showing the FD-optimal policy degrades gracefully under factor-of-two microstructure misspecification once the terminal-impact face-lift is implemented correctly. The resulting message is not that calibration always maximises cash premium, but that, inside a correct low-dimensional CJP model class, stronger modelling assumptions buy orders-of-magnitude gains in sample efficiency, value accuracy, and completion stability.

KEYWORDS

optimal execution; quasi-variational inequality; reinforcement learning; Hamilton–Jacobi–Bellman; stochastic intensity; sample complexity; limit order book; Cox–Ingersoll–Ross intensity

1. Introduction

Institutional liquidation of large positions trades off price uncertainty against execution cost. The discrete-time formulation of [Bertsimas and Lo \(1998\)](#) and the continuous-time analogue of [Almgren and Chriss \(2001\)](#) delivered closed-form deterministic schedules under linear impact; [Obizhaeva and Wang \(2013\)](#) extended this line to a stochas-

tic supply/demand process. The richer microstructure model of [Cartea et al. \(2015\)](#) (henceforth CJP) embedded the same objective into a Poisson-fill limit-order book and obtained a tractable quasi-variational inequality (QVI) that delivers the agent’s depth posting and market-order trigger jointly; related LOB-control and numerical-HJB treatments include [Bayraktar and Ludkovski \(2014\)](#), [Forsyth \(2011\)](#), and the book-length account of [Guéant \(2016\)](#). Subsequent CJP-style extensions cover model uncertainty ([Cartea et al., 2017, 2018](#)) and Hawkes-type self-excitation ([Alfonsi and Blanc, 2016](#)). For the empirical microstructure background we refer to the comprehensive treatments of [Bouchaud et al. \(2018\)](#) and [Cont et al. \(2010\)](#). The CJP reduction via the ansatz $H(t, x, S, q) = x + qS + h(t, q)$ exposes a one-dimensional dynamic-programming problem in h that admits closed-form solutions for small inventory and rapid numerical solution for general q via standard impulse-control numerical methods ([Forsyth and Vetzal, 2002](#); [Pham, 2009](#)).

Reinforcement learning has emerged as a popular data-driven alternative when microstructure parameters are unknown or non-stationary. [Nevmyvaka et al. \(2006\)](#) introduced Q-learning for execution scheduling; [Hendricks and Wilcox \(2014\)](#) added an RL layer to the [Almgren and Chriss \(2001\)](#) schedule; [Lin and Beling \(2020\)](#) applied end-to-end PPO to execution; [Ning et al. \(2021\)](#) trained a Double DQN on a CJP-like environment with raw limit-order-book features and reported a +6.5-bps premium over TWAP. Recent surveys ([Charpentier et al., 2023](#); [Hambly et al., 2023](#)) document the broader landscape of RL applications in quantitative finance. The promise of this line of work is that, by foregoing explicit microstructure modelling, RL controllers can adapt to environments where the CJP assumptions partially fail.

Our experiments suggest this promise should be qualified. In a CJP-faithful simulator at the CJP Chapter 8 scale, a Double DQN with $\sim 5\,000$ parameters saturates near a +0.129 dollar premium per 10-share liquidation episode and is dominated by a two-parameter MLE that attains the FD-optimal premium of about +0.194 from a small observed-fill sample. Equivalently, the plug-in premium is about 1.9 cents per share, or 6.5 basis points at the baseline midprice $S_0 = \$30$. The gap does not close with sample budget: at $n = 10^5$ environment transitions—two orders of magnitude beyond what is feasible on real high-frequency data within a calibration window—DDQN still trails the MLE. A structure-aware hybrid policy that hard-wires the CJP exponential ansatz $\hat{\kappa}^{-1}$ into the architecture closes part of the parameter-efficiency gap but not the premium gap at this scale, suggesting that the bottleneck is the model-free RL route rather than merely DDQN’s implementation.

Because the CJP Chapter 8 benchmark uses a deliberately small inventory, we also report a scale stress test with $Q_0 = 100$ and $T = 600$ s. Since both inventory and horizon increase by the same factor, the baseline fill intensity $\lambda = 50/60$ is kept fixed so that $\lambda T/Q_0$ is unchanged. The stress test changes the interpretation rather than simply amplifying the headline result: pure passive posting loses its apparent dominance once clearance is inspected, while some RL seed means tilt toward higher cash premium at the expense of value-function accuracy and completion. The plug-in calibrator remains the only method that both matches the FD solution and stays computationally negligible.

We make four contributions in support of this picture. First, we construct a reward shaping (Theorem 5.1) under which the expected cumulative return of any policy equals the reduced value function $h(t, q)$. This shaping promotes the Monte-Carlo premium comparison used by prior work to a pointwise diagnostic $\|V_\pi(t, q) - h(t, q)\|_\infty$ measurable on the FD reference grid. Second, we empirically characterise the sample-efficiency frontier between a strong parametric model-based planner and model-free

RL on the CJP Chapter 8 scale. An exact-Bernoulli MLE of $(\hat{\kappa}, \hat{\lambda})$ combined with one backward FD pass at the fitted parameters achieves the FD-optimal premium at every sample budget from 10^2 to 10^5 , while Tabular Q and DDQN remain below it. Third, we propose and evaluate a structure-aware hybrid policy class ($\delta_\theta = \hat{\kappa}^{-1} + g_\theta$, with g_θ a signed bounded residual MLP) trained via clipped Proximal Policy Optimization (Schulman et al., 2017). The Regime-II stress test shows why premium alone is not enough: the hybrid’s mean raw premium can be higher within seed-level noise, but it remains inaccurate and incomplete relative to the FD-matching calibrator. Fourth, we extend the framework to a CIR-type stochastic fill intensity (Lemma 6.1), derive the reduced HJB, and solve it with an upwind backward-Euler scheme. We validate the $\sigma_\lambda \rightarrow 0$ limit explicitly on $\sigma_\lambda \in [0, 0.1]$ at three intensity slices; the high-volatility regime is treated as a numerical extension rather than as a validation claim.

The remainder of the paper is structured as follows. Section 2 fixes notation and the model. Section 3 reviews the LO-only benchmark of Cartea and Jaimungal (2015). Section 4 treats LO+MO with the FD QVI. Section 5 sets up the four-agent benchmark and presents the structure-aware hybrid. Section 6 adds the CIR-intensity extension. Section 7 reports parameter robustness and misspecification. Section 8 extends the baseline panel. Section 9 discusses implications and limitations; Section 10 concludes.

2. Model and notation

We follow the discrete-time LO/MO model of Cartea et al. (2015, Chapter 8) on a fixed horizon T . The agent’s objective is to liquidate an initial inventory Q_0 shares by time T while balancing the LO fill premium against running inventory variance and a forced-liquidation cost. The baseline $Q_0 = 10$ is the small pedagogical inventory scale used in the CJP Chapter 8 numerical example; Section 9.3 discusses the scaling limitation. Unless otherwise stated, implementation premiums are reported in dollars per complete liquidation episode relative to TWAP. Dividing by Q_0 gives dollars per share. For the scale stress test in Section 8.1, we use $Q_0 = 100$, $T = 600\text{s}$, $S_0 = 30$, $\lambda = 50/60$, $\kappa = 100$, $\xi = 0.005$, $\alpha = 10^{-5}$, and $\phi = 10^{-6}$. Keeping λ fixed preserves the expected external-arrival density per share, $\lambda T/Q_0$, because Q_0 and T are both scaled by 10.

2.1. Notation

Table 1 lists every symbol used in the paper.

Definition 2.1 (Admissible controls). Let $\mathbb{F} = (\mathcal{F}_t)_{0 \leq t \leq T}$ be the natural filtration generated by the midprice Brownian motion, the external order-flow randomness, and the controlled counting processes. A control pair (δ, M) is admissible if δ is $\mathbb{F}$ -predictable with $0 \leq \delta_t \leq \delta_{\max} < \infty$, M is a simple $\mathbb{F}$ -adapted point process with unit jumps and $\mathbb{E}[M_T] < \infty$, and the induced inventory remains in $\{0, \dots, Q_0\}$, so M can jump only when $Q_{t-} > 0$. We denote this class by $\mathcal{A}$.

2.2. LO/MO dynamics

The midprice follows

$$dS_t = \sigma dW_t, \tag{1}$$

Table 1. Notation. All quantities are in seconds, dollars, and shares as appropriate.

Symbol	Meaning	Default value
T	trading horizon	60 s
Q_0	initial inventory	10 shares
q	remaining inventory	$0, \dots, Q_0$
δ	posted LO depth above the mid	≥ 0
S_t	midprice (arithmetic Brownian)	—
S_0	initial midprice	30 \$
σ	midprice volatility	0.01 $\$/\sqrt{s}$
λ	arrival rate of external MOs	50/60 s^{-1}
κ	depth-decay parameter	100 $\$^{-1}$
ξ	half-spread paid on agent MOs	0.005 \$
α	quadratic terminal-impact coefficient	10^{-3} $\$/share^2$
ϕ	running inventory-penalty coefficient	10^{-5} $\$/share^2/s$
$\tilde{\lambda}$	effective rate λ/e after optimising δ	—
$h(t, q)$	reduced value function from ansatz $H = x + qS + h$	—
$\omega(t, q)$	exponential transform $e^{\kappa h}$	—
κ_λ	CIR mean-reversion speed for λ_t	2.0
$\bar{\lambda}$	CIR long-run mean	50/60 s^{-1}
σ_λ	CIR vol-of-vol	0.1

with S_0 given. The driftless midprice is the CJP short-horizon assumption: over the 60-second liquidation window, directional drift is treated as negligible relative to execution costs and volatility. The agent posts a single unit-size LO at depth $\delta_t \geq 0$ above the prevailing mid. External MOs arrive as a Poisson process of rate λ ; conditional on an arrival, the agent's LO is hit with probability

$$P(\delta) = e^{-\kappa\delta}. \quad (2)$$

The agent may also fire a unit MO at any time, paying half the spread ξ per share. Inventory evolves as

$$dQ_t^{\delta, M} = -dN_t^\delta - dM_t, \quad (3)$$

where N^δ counts agent LO fills and M counts agent MOs. Cash accumulates as

$$dX_t^{\delta, M} = (S_{t-} + \delta_{t-}) dN_t^\delta + (S_{t-} - \xi) dM_t. \quad (4)$$

The agent's performance criterion is

$$H(t, x, S, q) = \operatorname{ess\,sup}_{(\delta, M) \in \mathcal{A}} \mathbb{E} \left[X_T + Q_T(S_T - \alpha Q_T) - \phi \int_t^T Q_s^2 ds \mid X_t = x, S_t = S, Q_t = q \right]. \quad (5)$$

The terminal expression marks any residual inventory to a block-liquidation value with quadratic impact; the agent can alternatively fire market orders immediately before maturity and pay the half-spread cost. This produces the face-lifted boundary

used below, $h(T^-, q) = \max\{-\alpha q^2, -q\xi\}$. The $-\phi \int_t^T Q_s^2 ds$ term penalises holding inventory during the trading interval; we use the predictable left-continuous inventory convention for the integrand, which is immaterial for finitely many jump times under Lebesgue integration.

2.3. Structural decomposition

Proposition 2.2 (Ansatz validity). *Under (1)–(5), the value function admits the additive separation*

$$H(t, x, S, q) = x + qS + h(t, q), \quad (6)$$

where h depends only on (t, q) and is bounded on $[0, T] \times \{0, \dots, Q_0\}$. In particular $h(t, 0) = 0$ for all $t \leq T$, $h(T, q) = -\alpha q^2$, and the pre-terminal face-lift is $h(T^-, q) = \max\{-\alpha q^2, -q\xi\}$.

Proof sketch. The midprice is a martingale, so $\mathbb{E}[qS_T - qS_t] = 0$ and the cash component x adds linearly. The face-lift follows by comparing the terminal block liquidation value $-\alpha q^2$ with firing q unit market orders immediately before T at total half-spread cost $q\xi$, as in [Cartea et al. \(2015, §8\)](#). $\square$

Throughout we work with the reduced function $h(t, q)$, and we use the exponential transform

$$\omega(t, q) = e^{\kappa h(t, q)} \quad (7)$$

freely; the LO-only solution (10) below is expressed most cleanly in ω .

3. Benchmark: limit-orders only

This section uses the LO-only degeneracy $\xi = \phi = 0$ as an analytical benchmark, not the LO+MO parameter regime used in Sections 4–8. When $\xi = \phi = 0$ the agent only ever posts LOs and the reduced HJB becomes the ODE system

$$\partial_t h(t, q) + \sup_{\delta} \lambda e^{-\kappa \delta} [\delta + h(t, q - 1) - h(t, q)] = 0, \quad h(t, 0) = 0, \quad (8)$$

with terminal $h(T, q) = -\alpha q^2$. In the LO-only regime ($\xi = \phi = 0$) the supremum is attained at the unique interior maximiser

$$\delta^*(t, q) = \kappa^{-1} + h(t, q) - h(t, q - 1). \quad (9)$$

In the LO+MO setting of Section 4, (9) remains the continuation-operator inner argmax—i.e., the optimal depth conditional on *not* firing an MO at the current step—and serves as the structural anchor for the hybrid policy of Section 5. Substituting (9) back into (8) gives the closed-form solution

$$\omega(t, q) = \sum_{n=0}^q \frac{\tilde{\lambda}^n}{n!} e^{-\kappa \alpha (q-n)^2} (T-t)^n, \quad \tilde{\lambda} = \lambda/e, \quad (10)$$

of [Cartea et al. \(2015, §8.2\)](#). The optimal depth has two interpretable terms: a κ^{-1} “asymptote” equal to the depth that maximises the per-fill premium $\delta e^{-\kappa\delta}$, and a log-likelihood-ratio correction $h(t, q) - h(t, q - 1)$ that captures the marginal continuation value of holding one fewer share. The asymptote κ^{-1} is the structural prior we exploit in the hybrid policy of Section 5.

The LO-only formula is used only to motivate the depth asymptote and the CJP exponential transform. The full sample-complexity comparison appears in Section 5.3, where the market-order action, half-spread cost, running inventory penalty, and face-lift boundary are all active.

4. Mixed control: limit and market orders

When $\xi > 0$ and $\phi > 0$ the agent has a non-trivial choice between waiting for an LO fill and crossing the spread with an MO. The reduced QVI of [Cartea and Jaimungal \(2015\)](#) is

$$\max \left\{ \partial_t h - \phi q^2 + \frac{\tilde{\lambda}}{\kappa} e^{-\kappa[h(t, q) - h(t, q-1)]}; \right. \\ \left. -\xi + h(t, q-1) - h(t, q) \right\} = 0, \quad (11)$$

with $h(t, 0) = 0$ and the face-lifted terminal $h(T^-, q) = \max\{-\alpha q^2, -q\xi\}$. The two arguments of max correspond to continuation (post an LO) and intervention (fire an MO), respectively; the QVI is active where the intervention argument matches the continuation argument, defining the MO trigger boundary.

Theorem 4.1 (QVI well-posedness). *Under Assumption $\xi, \alpha, \phi \geq 0$ and the running-cost integrability of (5), the QVI (11) admits a unique bounded solution $h : [0, T] \times \{0, \dots, Q_0\} \rightarrow \mathbb{R}$, continuous in t and satisfying the stated boundary and face-lift terminal conditions. The associated optimal policy is feedback in (t, q) .*

Proof reference. The QVI (11) is the dynamic-programming equation of a finite-state impulse-control problem with state $(t, q) \in [0, T] \times \{0, \dots, Q_0\}$, inventory-only impulse ($q \mapsto q - 1$ at cost ξ), and continuous control $\delta \in [0, \delta_{\max}]$. The finite-state standard result of [Pham \(2009, §5.4\)](#) applies once the continuation operator is locally Lipschitz in the finite set of inventory coordinates. An independent a priori bound follows by comparison with immediate market-order liquidation and with the bounded terminal payoff: for all q , $-q\xi - \phi q^2 T - \alpha q^2 \leq h(t, q) \leq \delta_{\max} q$. Hence $|\Delta_q h|$ is bounded by a constant depending only on $(Q_0, \xi, \alpha, \phi, T, \delta_{\max})$, and $\sup_{\delta} \lambda e^{-\kappa\delta} [\delta + \Delta_q h]$ is locally Lipschitz on the resulting compact range. The feedback structure of the optimal policy is standard for finite-state impulse problems; comparison-principle arguments such as [Bouchard and Touzi \(2011, Theorem 4.1\)](#) give the analogous uniqueness statement in continuous spatial domains. $\square$

4.1. Analytical solution for $q \in \{1, 2\}$

For orientation, consider the spread-face-lift subcase in which the terminal branch is $h(T^-, q) = -q\xi$. This is the closed-form calculation in CJP and is useful for reading the free-boundary geometry, while the numerical experiments below use the full ter-

minimal condition $\max\{-\alpha q^2, -q\xi\}$. For $q = 1$ in the continuation regime $\phi \leq \tilde{\lambda}e^{\kappa\xi}/\kappa$, integrating the QVI gives

$$\omega(t, 1) = e^{-\kappa\xi}e^{-\kappa\phi(T-t)} + \frac{\tilde{\lambda}}{\kappa\phi}(1 - e^{-\kappa\phi(T-t)}). \quad (12)$$

For the main-regime parameters, $\phi/(\tilde{\lambda}e^{\kappa\xi}/\kappa) \approx 2 \times 10^{-3} \ll 1$, so this continuation-regime condition is comfortably satisfied. For $q = 2$ there is a critical time $\tau_2 \in (0, T]$ at which the MO trigger first activates; on $[0, \tau_2)$ the continuation ODE applies, and on $[\tau_2, T]$ the trigger is on and $\omega(t, 2) = e^{-\kappa\xi}\omega(t, 1)$. The critical time solves

$$\omega(\tau_2, 1) = \frac{\tilde{\lambda}e^{\kappa\xi}}{\tilde{\lambda} - 3\kappa\phi e^{-\kappa\xi}}. \quad (13)$$

4.2. Finite-difference scheme

For general q we discretise time uniformly on $[0, T]$ with step Δt and solve (11) backward in time with explicit Euler. At each step we compute the continuation value

$$h_{\text{cont}}(t, q) = h(t + \Delta t, q) - \Delta t \cdot \left(\phi q^2 - \frac{\tilde{\lambda}}{\kappa} e^{-\kappa[h(t+\Delta t, q) - h(t+\Delta t, q-1)]} \right), \quad (14)$$

the intervention value $h_{\text{exec}}(t, q) = h(t, q - 1) - \xi$ (using the just-updated h at $q - 1$), and set $h(t, q) = \max\{h_{\text{cont}}, h_{\text{exec}}\}$. The MO trigger at (t, q) records whether the intervention argument wins.

Proposition 4.2 (FD stability). *The explicit Euler scheme (14) is monotone and consistent. It is stable whenever $\Delta t \cdot \frac{\tilde{\lambda}}{\kappa} e^{\kappa d_{\max}} < 1$, where $d_{\max}$ bounds the depth iterates. Monotonicity and consistency, together with the comparison principle for the finite-state QVI, give convergence in the sense of the Barles–Souganidis (Barles and Souganidis, 1991) viscosity-solution program, here applied at the discrete-inventory level.*

For the book parameters $\lambda = 50/60$, $\kappa = 100$, $d_{\max} \approx 0.03$ we use $\Delta t = 0.01$ which is comfortably stable. The FD solution at these parameters is the analytical-reference grid used throughout the paper.

5. Structure-aware reinforcement learning

We now construct a four-agent benchmark on the same simulator. The motivating question is: which design choices in an RL agent for execution *actually* reduce sample complexity, and which simply add free parameters?

5.1. Reward-shaped MDP

Because S is a martingale and the value separates as $H = x + qS + h$, the only components of the change in H over a single environment step that survive expectation

are the LO fill premium, the MO half-spread cost, the running inventory penalty, and the terminal face-lift. This motivates the following shaping.

Theorem 5.1 (Reward shaping yields $V \equiv h$). *Define the per-step reward*

$$r_t = -\phi Q_t^2 \Delta t - \xi \cdot \mathbf{1}_{\{MO \text{ fired at } t\}} + \delta_t \cdot \mathbf{1}_{\{LO \text{ filled at } t\}} - \min\{\alpha Q_T^2, \xi Q_T\} \cdot \mathbf{1}_{\{t=T, Q_T > 0\}}. \quad (15)$$

Then under any policy π , the expected cumulative reward $J^\pi(t, q) = \mathbb{E}_\pi \left[\sum_{s=t}^T r_s \mid Q_t = q \right]$ satisfies $J^\pi(t, q) = h^\pi(t, q)$, where h^π is the reduced value function under π . Consequently, $J^{\pi^}(t, q) = h(t, q)$ with h the solution of (11). The identity is exact in continuous time; for the discrete-time MDP used in simulation it holds up to the same $O(\Delta t)$ truncation error as the explicit-Euler discretisation in Section 4.2.*

Proof sketch. A telescoping argument on $H_t - H_s$ along the controlled state under (3)–(4) cancels all martingale contributions from S ; what remains coincides term-by-term with (15). Details are in Appendix A. $\square$

Theorem 5.1 promotes “ $\|V_{\text{RL}} - h\|_\infty$ ” to a meaningful pointwise diagnostic. Tabular Q, Double DQN, the structure-aware hybrid, and the plug-in MLE estimator are all evaluated against the same FD-derived h grid.

5.2. Four agents

Agent A — Tabular Q. A $\mathcal{O}(|S||A|)$ -parameter tabular Q-learner on state (t_{idx}, q) with discretised time bucket count 121, inventory cardinality $Q_0 + 1 = 11$, and 22 discrete actions (21 depth grid points + MO). The baseline run uses ϵ -greedy exploration only; learning rate is annealed from 0.3 to 0.05.

Agent B — Double DQN. A standard van Hasselt et al. (2016) Double DQN with a two-layer 64-hidden-unit MLP, replay buffer of 5×10^4 , target sync every 500 steps, batch size 128, Adam at 10^{-3} . Input features $[t_{\text{idx}}/n_t, q/Q_0]$. The baseline run again uses ϵ -greedy exploration only.

Agent C — Structure-aware hybrid. The novel contribution. A trunk with two hidden layers of 16 units each feeds three heads (detailed in Appendix B):

- $g_\theta(t, q) \in [-0.01, 0.04]$: signed residual depth correction. Posted depth is $\delta_\theta = \kappa^{-1} + g_\theta$, then snapped to the admissible depth grid. The signed residual allows depths below κ^{-1} , including the zero-depth grid point.
- $h_\theta(t, q) \in \mathbb{R}$: value head. Serves as the actor-critic baseline in the PPO loss and as the input to the MO trigger logit.
- $\log \beta_\theta \in \mathbb{R}$: a learned scalar temperature for the MO gate, $\pi_{\text{MO}}(t, q) = \sigma(\beta_\theta \cdot [(h_\theta(t, q - 1) - h_\theta(t, q)) - \xi])$.

Total parameter count ≈ 350 . Training uses clipped PPO (Schulman et al., 2017) with 4 epochs per batch, advantage normalisation, gradient clip 1.0, value-loss coefficient 5.0, Adam at 3×10^{-4} . Per-step continuous depth $\delta_\theta + \mathcal{N}(0, \sigma_{\text{explore}})$ is snapped to the env depth grid via a categorical Gaussian-density softmax for log-prob differentiability. Because the MO gate is cold-started near zero, the training rollout is additionally mixed with a forced MO-exploration probability decaying linearly from 5% to zero; this is an exploration injection, not standard PPO.

Table 2. Four-agent comparison at the largest sample budget. Premium values are mean $\pm$ standard deviation across 20 seeds; wall-clock is mean training time per seed on a single CPU core.

Agent	val_err RMSE @ 10^4	premium @ 10^4	premium @ 10^5	wall (s)
A. Tabular Q	0.063	$+0.096 \pm 0.007$	$+0.134 \pm 0.007$	0.4
B. DDQN	0.060	$+0.131 \pm 0.050$	$+0.129 \pm 0.057$	70.9
C. Hybrid (PPO)	0.118	$+0.082 \pm 0.075$	$+0.135 \pm 0.048$	23.8
D. Plug-in MLE	0.003	$+0.192 \pm 0.006$	$+0.194 \pm 0.007$	0.3

Agent D — Plug-in MLE. A pure model-based calibrator. Run a random-depth exploration policy until n_{fills} fills have been observed, then jointly maximise the exact Bernoulli log-likelihood $\ell(\lambda, \kappa) = \sum_i [y_i \log(1 - e^{-\eta_i}) - (1 - y_i) \eta_i]$, $\eta_i = \lambda e^{-\kappa \delta_i} \Delta t$, under a weak log-Gaussian prior on $(\log \kappa, \log \lambda)$ via L-BFGS-B. The fitted $(\hat{\kappa}, \hat{\lambda})$ are plugged into a fresh FD solve, and the FD policy is deployed. No gradient updates on a neural network.

Connection to model-based RL. Agent D is not an oracle and not a model-free learner; it is the parametric model-based endpoint of the comparison in the sense of classical planning with a learned model (Bertsekas, 2019; Sutton and Barto, 2018). It assumes the CJP fill family is correct, estimates its two unknown parameters, and then plans exactly inside that fitted model. A fully nonparametric model-based RL agent would instead learn transition dynamics and then plan, while Agents A–C learn value or policy objects directly from rewards. Thus the benchmark’s main axis is not “statistics versus learning” in the abstract, but assumption strength versus sample efficiency: when the low-dimensional CJP model class is correct, the parametric planner is the sample-efficient limit; when that model class fails, the advantage can plausibly shift toward richer learned dynamics or policies.

For the value- and policy-error diagnostics below, each agent’s environment-time grid is compared with the FD reference by nearest-neighbour lookup in time and exact lookup in inventory; no smoothing over the inventory coordinate is used.

5.3. Sample-complexity benchmark

We sweep four agents over seven sample budgets $n \in \{10^2, 10^{2.5}, 10^3, 10^{3.5}, 10^4, 10^{4.5}, 10^5\}$ environment transitions and twenty independent seeds per cell (560 runs total). Each cell trains the agent, computes the value error $\|V_{\text{RL}} - h_{\text{FD}}\|_{\text{RMSE}}$ on the env grid, the mean policy error $|\delta_{\text{RL}} - \delta^*|_{L^1}$, and the implementation premium versus a TWAP baseline over 2000 Monte-Carlo evaluation paths with paired-bootstrap (BCa) 95% confidence intervals.

Figure 1 shows the resulting four-panel sample complexity diagnostic. The vertical bars are standard errors of the mean across the twenty seeds.

The headline numbers at $n = 10^5$ appear in Table 2.

Three observations. First, Agent D dominates the premium and value-error metrics at every budget. Once the agent’s model class is correct (the CJP environment), a ~ 100 -fill MLE plus one FD solve already attains a value-RMSE near the noise floor of the FD scheme itself. Second, Agent A (Tabular) shows the cleanest sample-complexity curve, with premium climbing to about $+0.134$ at $n = 10^5$. Third, Agent C (Hybrid PPO) becomes competitive with the model-free RL agents after the signed-residual

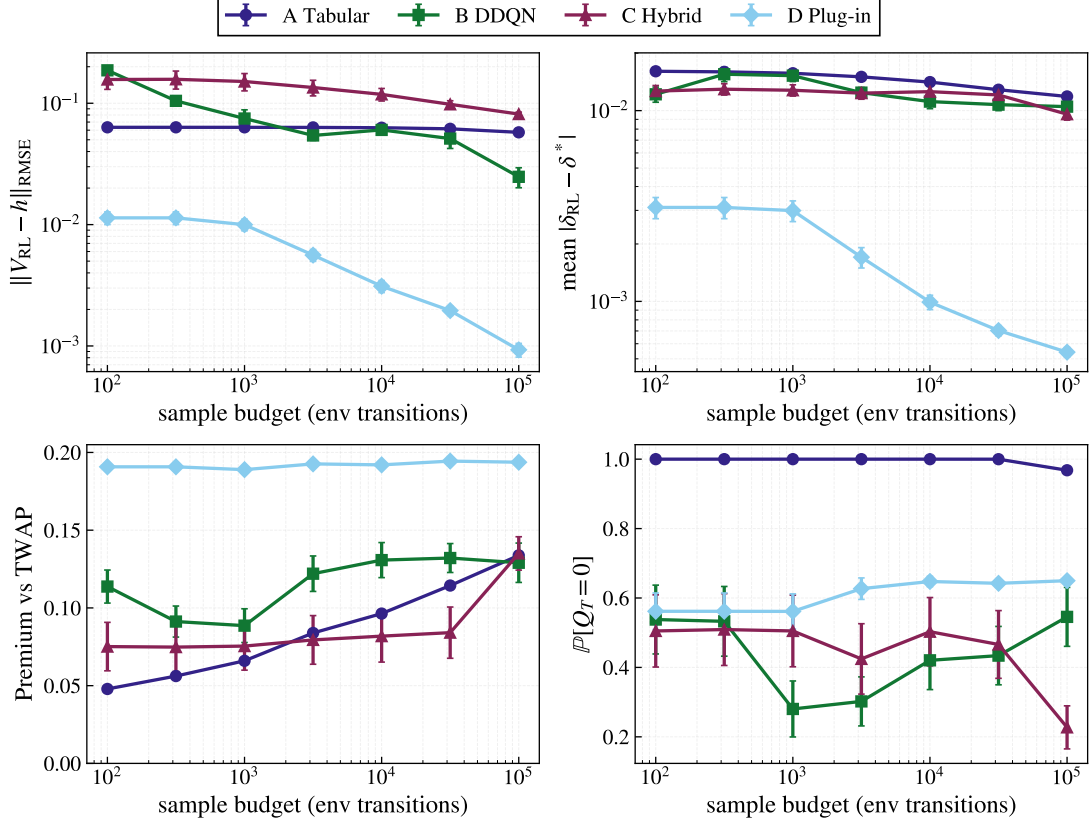


Figure 1. Sample-complexity benchmark in the main LO+MO parameter regime, mean $\pm$ SEM over 20 independent seeds per cell. Panel (a): RMSE of the learned value function against the FD reference h . Panel (b): mean absolute policy error $|\delta_{\text{RL}} - \delta^*|$. Panel (c): implementation premium versus TWAP, in dollars per 10-share liquidation episode, across 2000 MC paths per cell. Panel (d): probability that the agent fully clears its inventory by T . Agent D (plug-in MLE, $\diamond$) attains essentially the FD-optimal premium at every budget; Agents A–C learn the policy from rewards alone and only catch up partially even at $n = 10^5$.

architecture fix, but it still exhibits high run-to-run variance and remains below the plug-in premium.

We note that DDQN’s premium saturates near $+0.130$ starting at $n = 10^{3.5}$ and has highly variable clearance behaviour across seeds. At $n = 10^5$ its mean clearance probability is only about 0.55 with a seed standard deviation near 0.38, which is a stronger instability signal than the premium curve alone. This behaviour is consistent with replay-buffer overwriting and target-network drift artefacts documented in [van Hasselt et al. \(2016, §5\)](#); our replay buffer of 5×10^4 transitions is smaller than $n = 10^5$, so older transitions are overwritten and the effective training distribution is shifted toward the policy’s late-stage behaviour. We did not pursue further hyperparameter tuning as DDQN’s ceiling is well below the plug-in benchmark regardless.

The premium gap between every RL agent and the plug-in calibrator is the principal empirical result of the paper. We return to its interpretation in Section 9.

5.4. Failure mode diagnostics

Figure 2 reports two structural failure modes diagnosed during the development of the four-agent benchmark.

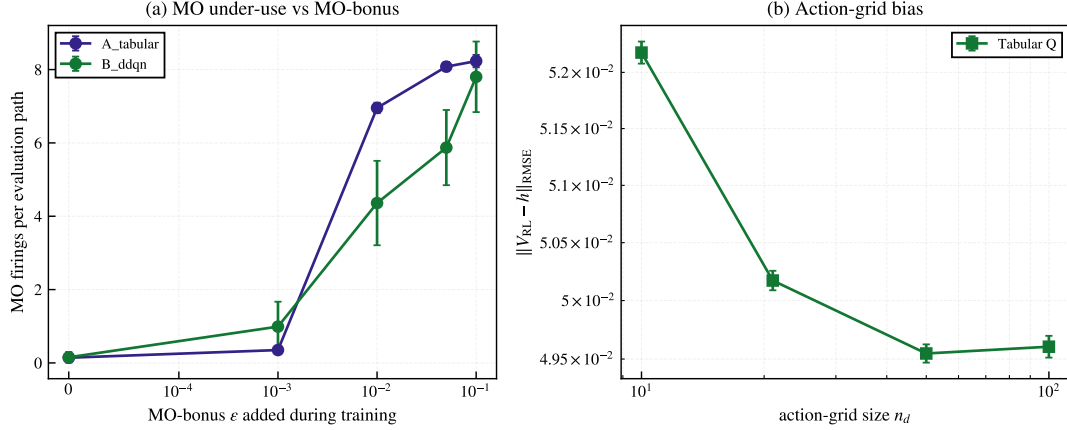


Figure 2. RL failure-mode diagnostics. Panel (a): MO firings per evaluation path as a function of an exploration ϵ -bonus added to the MO action’s reward during training only (evaluation uses the canonical reward of Theorem 5.1). At $\epsilon = 0$, DDQN and Tabular Q both under-use MOs relative to the FD trigger, although the DDQN outcome is seed-sensitive. As the training bonus rises, both agents shift toward heavier MO use, confirming that the under-use at $\epsilon = 0$ is exploration-bias driven rather than a genuine optimum. Panel (b): RMSE of the learned value function against h as a function of the depth-action grid size n_d . Tabular Q’s value error decreases only mildly as n_d increases, showing that grid resolution is a secondary limiter rather than the sole source of value error.

Table 3. Forced-MO exploration fairness ablation for Agents A and B. Runs use the Regime-I parameter set, 20 training seeds, a 10^5 -transition-equivalent budget (826 training episodes), and 500 Monte-Carlo evaluation paths per seed. Premium is mean dollars per 10-share episode versus TWAP; “s.e.” is the seed-level standard error. Mean premiums differ slightly from Table 2 because this ablation uses 500 evaluation paths per seed whereas Table 2 uses 2000 paths per cell; the difference is within Monte-Carlo standard error.

Agent	Exploration	Premium	s.e.	Clearance	MO/path
Tabular Q	baseline ϵ -greedy	+0.140	0.004	0.969	0.055
Tabular Q	forced MO 5% $\rightarrow$ 0	+0.135	0.004	1.000	0.119
DDQN	baseline ϵ -greedy	+0.133	0.012	0.540	0.028
DDQN	forced MO 5% $\rightarrow$ 0	+0.136	0.010	0.466	0.350

Panel (a) is the most consequential observation: agents trained on the canonical CJP reward under-sample the rare MO action under ordinary ϵ -greedy exploration. Adding even a tiny per-MO bonus to the training reward (Tabular: $\epsilon \in \{10^{-3}, 10^{-2}\}$ already lifts MO use; DDQN: similar but noisier shift) demonstrates that the under-use is an exploration artefact, not a learned optimum. Hybrid Agent C uses a different exploration scheme: its MO trigger is driven by the analytic identity $h(t, q - 1) - h(t, q) > \xi$ wired in through the gate logit, and the rollout policy is augmented by a forced MO-exploration probability that decays from 5% to zero. To make this privilege explicit, Table 3 gives Agents A and B the same decaying forced-MO hook used by Agent C.

The fairness ablation does not close the calibration gap. For Tabular Q, forced MO exploration improves terminal clearance but reduces premium slightly because extra crossings give up spread. For DDQN, the hook raises MO usage and improves premium only from +0.133 to +0.136, far below the plug-in MLE’s +0.194 in Table 2. Its clearance falls from 0.540 to 0.466 despite the higher MO count because the injected exploration distribution shifts the learned terminal-trigger timing; this is another sign of DDQN’s sensitivity to the exploration distribution, not an improvement in the

Table 4. Summary of RL failure modes diagnosed in this paper and the agent classes they affect. *Affected* = the agent class for which the failure is empirically active at default hyperparameters. *Mechanism* identifies the structural reason. *Mitigation* describes the patch deployed in this paper (where applicable).

Failure mode	Affected	Mechanism	Mitigation
MO under-use	DDQN, Tabular Q	ϵ -greedy exploration starves the rare MO action	MO-action bonus during training; evaluation reward unchanged
Action-grid bias	Tabular Q	discrete depth grid has finite resolution	increase n_d , or move to continuous-action methods such as Hybrid PPO
MO-gate cold start	Hybrid PPO	h_θ initialised near zero gives $p_{\text{MO}}^{\text{net}} \approx \sigma(-\beta\xi) \approx 0$ at training start; the gate rarely fires and the agent collects little MO experience	forced 5% MO-exploration probability mixed into the gate, decaying linearly to zero; A/B fairness ablation reported separately

LO/MO trade-off. Hybrid PPO’s own endpoint clearance (0.227 in Table 6) should be read in the same diagnostic spirit: wiring the analytic trigger into the gate logit does not by itself make the rare terminal MO decision reliable when the critic head that drives the trigger is still noisy. Thus Hybrid PPO’s exploration privilege is a real design difference, but it does not explain the main plug-in-versus-RL premium gap. Panel (b) shows that action-grid resolution contributes only a small part of Tabular Q’s value error: increasing n_d from 10 to 100 changes RMSE by roughly 2×10^{-3} , so the dominant error is not merely depth discretisation.

Table 4 summarises the failure modes identified in this section and the agent classes affected.

6. Stochastic intensity extension

Real fill rates fluctuate. We extend the CJP framework to a CIR-type stochastic intensity (Andersen, 2008; Cox et al., 1985)

$$d\lambda_t = \kappa_\lambda(\bar{\lambda} - \lambda_t) dt + \sigma_\lambda \sqrt{\lambda_t} dW_t^\lambda, \quad (16)$$

with W^λ independent of the midprice Brownian motion. The Feller condition $2\kappa_\lambda\bar{\lambda} \geq \sigma_\lambda^2$ ensures $\lambda_t > 0$ almost surely; our default parameters $\kappa_\lambda = 2.0$, $\bar{\lambda} = 50/60$, $\sigma_\lambda = 0.1$ satisfy $2 \cdot 2 \cdot 50/60 \approx 3.33$, far larger than $\sigma_\lambda^2 = 0.01$.

6.1. HJB on the augmented state

The state is now (t, x, S, q, λ) . The ansatz still simplifies the problem:

Lemma 6.1 (Ansatz under CIR intensity). *Under (1)–(5) with λ_t replaced by the CIR process (16), the value function admits $H(t, x, S, q, \lambda) = x + qS + h(t, q, \lambda)$, with $h(t, 0, \lambda) = 0$ for all $\lambda > 0$ and the same face-lifted terminal $h(T^-, q, \lambda) =$*

$\max\{-\alpha q^2, -q\xi\}$. The reduced function $h(t, q, \lambda)$ solves the QVI

$$\max\left\{\partial_t h + \mathcal{L}_\lambda h - \phi q^2 + \frac{\tilde{\lambda}}{\kappa} e^{-\kappa[h(t,q,\lambda) - h(t,q-1,\lambda)]}; \right. \\ \left. - \xi + h(t, q-1, \lambda) - h(t, q, \lambda)\right\} = 0, \quad (17)$$

where here $\tilde{\lambda} = \lambda/e$ at the current intensity and $\mathcal{L}_\lambda = \kappa_\lambda(\bar{\lambda} - \lambda)\partial_\lambda + \frac{1}{2}\sigma_\lambda^2\lambda\partial_{\lambda\lambda}^2$ is the CIR infinitesimal generator.

Proof reference. Apply Ito's lemma to H along the controlled (X, S, Q, λ) process; the λ -diffusion adds the $\mathcal{L}_\lambda h$ term to the HJB. Independence of W and W^λ implies the additive separability is preserved. See Appendix A. $\square$

6.2. Numerical scheme: upwind backward Euler

We discretise λ log-uniformly on $[0.5\bar{\lambda}, 2\bar{\lambda}]$ and augment the grid with the three validation slices $0.7\bar{\lambda}$, $\bar{\lambda}$, and $1.3\bar{\lambda}$; t is uniform with $\Delta t = 0.05$. Each backward time step first applies an implicit backward-Euler solve for the CIR generator in λ , using a generator-form upwind stencil for the mean-reversion drift and central differences for the diffusion on the non-uniform grid. It then applies the QVI continuation/intervention update explicitly in the inventory direction. This is a first-order Lie–Trotter splitting, so the overall time accuracy is $O(\Delta t)$. Neumann boundary conditions are imposed at $\lambda_{\min}, \lambda_{\max}$; for the validated range $\sigma_\lambda \leq 0.1$, the stationary CIR distribution has essentially all of its mass inside $[0.5\bar{\lambda}, 2\bar{\lambda}]$, making boundary effects negligible at the reported precision. This corrected upwind direction is essential: the earlier central/incorrect-upwind variant produced non-physical large gaps at small positive σ_λ . With $\sigma_\lambda = 0$ and $\kappa_\lambda = 0$, the continuous-time problem reduces to the constant- λ FD of Section 4.2; numerically the mean slice matches that benchmark to the tolerance reported below.

6.3. Numerical results

Figure 3 shows three $h(t=0, q, \lambda)$ panels for $\sigma_\lambda \in \{0.0, 0.05, 0.10\}$ on a shared colour scale. The continuation value increases monotonically with λ at every fixed q , reflecting that higher fill intensity makes LO posting more attractive; the three panels are visually almost indistinguishable at these moderate σ_λ , consistent with the $O(\sigma_\lambda^2)$ leading-order perturbation visible in Figure 4 (next subsection).

In the main LO+MO parameter regime the MO-trigger geometry collapses to an essentially one-dimensional pattern: the intervention region is active for all $q \geq 2$ at every λ and is non-trivial only on the $q = 1$ line. A two-dimensional phase diagram therefore adds little information beyond the analytic $q \in \{1, 2\}$ result of Section 4.1. The validation below restricts attention to $\sigma_\lambda \leq 0.1$, where the small-volatility limit is resolved cleanly.

6.4. Numerical validation: $\sigma_\lambda \rightarrow 0$ limit

Figure 4 reports a multi-slice volatility validation at $\lambda_0 \in \{0.7\bar{\lambda}, \bar{\lambda}, 1.3\bar{\lambda}\}$. At the mean slice, $\sigma_\lambda = 0$ coincides with the constant- λ problem; away from the mean, deterministic mean reversion is already present at $\sigma_\lambda = 0$, so the plotted diagnostic is $\|h_{\text{CIR}}^\sigma(\cdot, \lambda_0, \cdot) -$

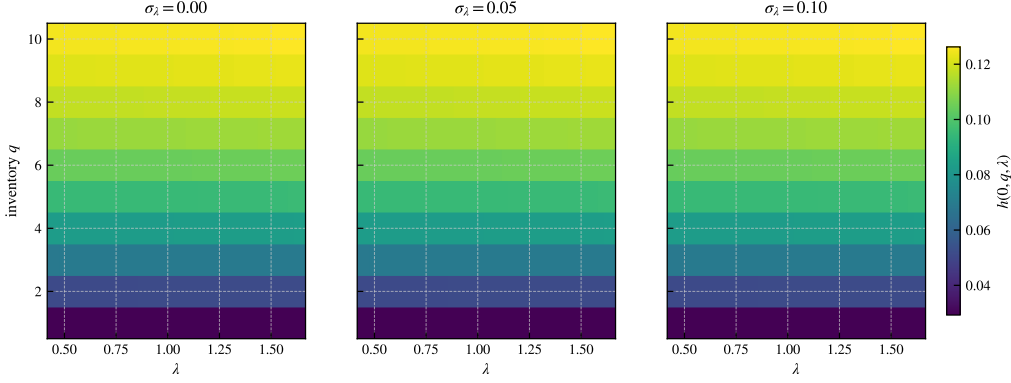


Figure 3. Reduced value function $h(0, q, \lambda)$ under CIR-intensity dynamics for $\sigma_\lambda \in \{0, 0.05, 0.10\}$. The three panels share a single colour scale; the visual near-identity reflects that the leading-order stochastic-intensity correction to h is small in the validated small-volatility regime. The $\sigma_\lambda = 0$ panel is the deterministic-intensity sanity check.

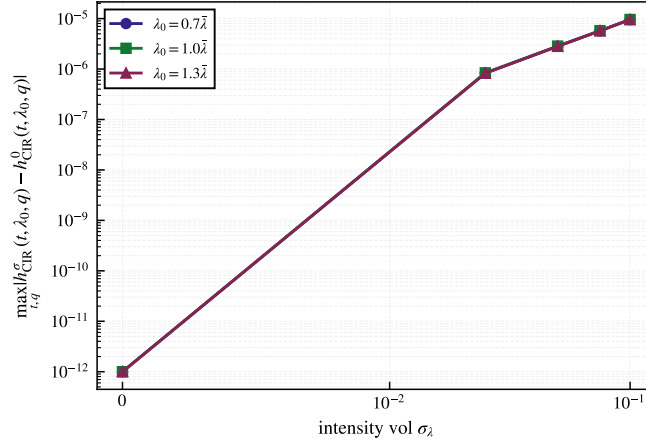


Figure 4. CIR scheme validation at three initial intensity slices. The plotted quantity is $\|h_{\text{CIR}}^\sigma(\cdot, \lambda_0, \cdot) - h_{\text{CIR}}^0(\cdot, \lambda_0, \cdot)\|_\infty$ for $\lambda_0 \in \{0.7\bar{\lambda}, \bar{\lambda}, 1.3\bar{\lambda}\}$. At the mean slice the $\sigma_\lambda = 0$ reference also coincides with the constant- λ FD solution. All positive-volatility gaps remain below 10^{-5} through $\sigma_\lambda = 0.1$.

$h_{\text{CIR}}^0(\cdot, \lambda_0, \cdot)\|_\infty$ at the same initial intensity slice. Across all three slices, the gaps at $\sigma_\lambda \in \{0.025, 0.05, 0.075, 0.10\}$ are approximately 8.3×10^{-7} , 2.8×10^{-6} , 5.7×10^{-6} , and 9.6×10^{-6} , respectively. Relative to $\|h_{\text{CIR}}^0\|_\infty \approx 0.126$, these are below 8×10^{-5} . The growth is approximately quadratic in σ_λ , consistent with the leading-order $O(\sigma_\lambda^2)$ perturbation of the deterministic-intensity ODE. This validates the small-volatility perturbation limit on and off the long-run mean; it is not a claim that the same coarse grid is validated for arbitrary high intensity volatility.

7. Parameter robustness and misspecification

We sweep each microstructure parameter over multipliers $\{0.5, 0.75, 1.0, 1.5, 2.0\}$ of its main-regime baseline value, recompute the FD-optimal policy, and evaluate the resulting implementation premium against TWAP over 2000 Monte-Carlo paths. Table 5 reports the univariate sweeps.

Table 5. Univariate parameter sensitivity. Each row varies one parameter while holding the rest at the main-regime defaults. Premium is FD-optimal versus TWAP, in dollars per 10-share liquidation episode, over 2000 MC paths using the robustness-script seed; small differences from the extended-baseline table reflect Monte-Carlo path-set variation.

Parameter \ multiplier	0.5	0.75	1.0	1.5	2.0
λ (fill rate)	+0.136	+0.180	+0.201	+0.242	+0.271
κ (depth decay)	+0.357	+0.250	+0.201	+0.161	+0.127
ξ (half-spread)	+0.173	+0.188	+0.201	+0.226	+0.251
α (terminal impact)	+0.200	+0.205	+0.201	+0.203	+0.198
ϕ (running penalty)	+0.215	+0.203	+0.201	+0.207	+0.205

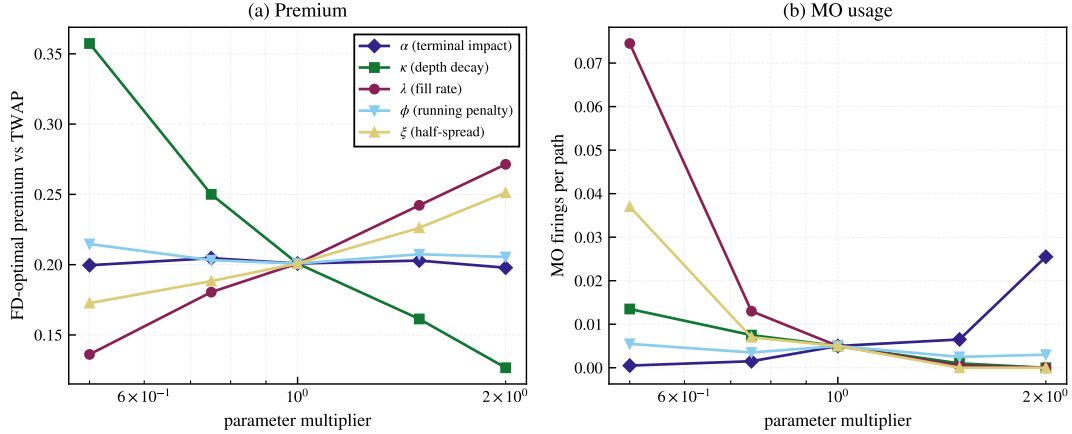


Figure 5. Univariate parameter sensitivity. Panel (a): FD-optimal premium vs TWAP under each of the five microstructure parameters being scaled separately by $\{0.5, 0.75, 1, 1.5, 2\} \times$ its baseline. Panel (b): the corresponding MO-firing rate per path. The λ and κ sweeps produce the steepest premium responses; α and ϕ are weaker but no longer exactly invariant after the corrected terminal face-lift.

Economically the main directions are correct: higher λ raises the LO premium (the agent fills faster); higher κ erodes it (the depth-premium trade-off becomes less attractive); higher ξ raises both TWAP cost and FD cost but the FD policy avoids MOs more, widening the gap. The corrected face-lift removes the spurious exact invariance in the α row: α now has a small but visible effect over factor-of-two perturbations. The small non-monotone movement in that row is within per-cell Monte-Carlo error and reflects that, at this scale, α is often dominated by the spread branch of the face-lift. An extreme stress test gives premiums +0.208, +0.202, +0.185, and +0.185 for $\alpha \in \{10^{-6}, 10^{-3}, 1, 10^3\}$, respectively, confirming that α is passed through the FD solver and Monte-Carlo evaluator. The high- α near-invariance is expected rather than suspicious: once $\alpha q^2 \geq q\xi$, the terminal face-lift saturates at the spread branch $-q\xi$. Figure 5 visualises the same sweeps as line plots, alongside the corresponding MO usage. With the corrected terminal face-lift, the FD policy uses MOs sparingly in the baseline regime; most of the premium response comes from the LO fill probability and the terminal face-lift trade-off rather than heavy MO intervention.

We additionally ran the two bivariate sweeps (κ, ϕ) and (ξ, α) over the same 5×5 multiplier grid; the resulting premium surfaces are dominated by single-parameter sensitivities and the numerical heatmaps are reported in Appendix C for completeness.

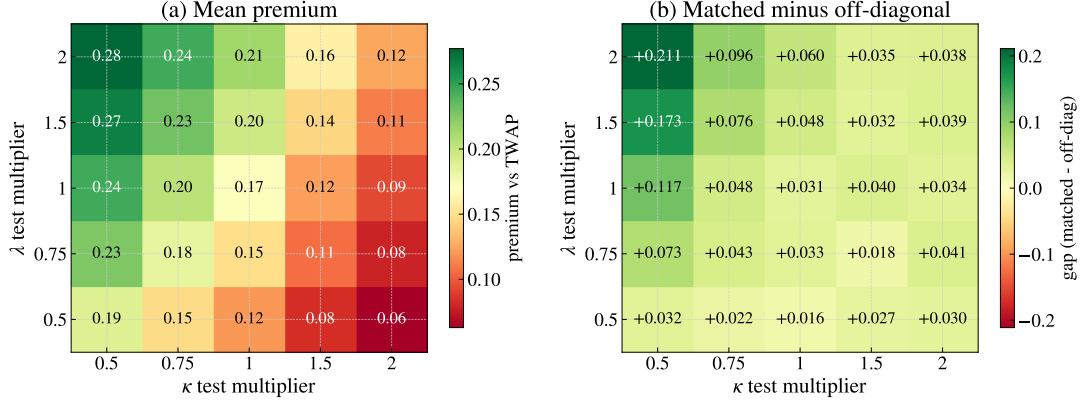


Figure 6. Train-test microstructure-parameter misspecification on (λ, κ) . Panel (a): mean premium across the 5×5 training grid for each test cell; the entire grid is in the green/positive regime. Panel (b): matched-training premium minus off-diagonal-mean premium per test cell—the per-cell cost of using the wrong calibration. Cell labels are numerical premiums; the colour scale is centred at zero on the same range in both panels.

The (ξ, α) surface now shows small but nonzero movement along the α direction, as it should.

We turn instead to the most operationally relevant experiment, summarised in Figure 6: a $5 \times 5 \times 5 \times 5 = 625$ -cell train/test misspecification grid. We train the FD-optimal policy at $(\lambda_{\text{train}}, \kappa_{\text{train}})$ and deploy it at $(\lambda_{\text{test}}, \kappa_{\text{test}})$. Panel (a) of the figure averages the realised premium across all training cells for each test cell—this is the “what does this test cell look like if I picked a random training calibration?” view. Panel (b) reports the more informative quantity for an execution desk: the matched-training premium (train = test) minus the mean off-diagonal premium (train $\neq$ test), per cell. A positive entry means there is a genuine penalty for mis-calibration at that test cell; a near-zero entry means the FD-optimal policy is essentially calibration-insensitive there.

Three observations from Figure 6. First, Panel (a) shows that every test cell sits in the positive-premium regime: the FD-optimal policy degrades gracefully under microstructure-parameter misspecification of factor ≤ 2 in either direction. Second, Panel (b) shows that the calibration sensitivity is asymmetric: cells with large λ_{test} and small κ_{test} (top-left quadrant) carry the largest matched-training advantage, while the diagonal cells of low-low and high-high parameters are nearly insensitive. Third, this robustness is the empirical underpinning of the plug-in agent’s success in Section 5: even a noisy MLE on a few hundred fills lands inside the green regime of Panel (a), so the resulting deployed policy keeps most of the FD-optimal premium.

8. Extended baselines and statistical rigour

We compare the FD-optimal and four RL agents against a panel of execution benchmarks: TWAP, Almgren–Chriss (Almgren and Chriss, 2001), VWAP (synthetic U-shape intraday volume), POV at 10%, a pure-passive LO at fixed depth $2/\kappa$, and a pure-aggressive policy. All baselines use the simulator-compatible interface $\pi(t, q, S) \rightarrow (\delta, \text{MO flag})$. TWAP here is the simulator-faithful constant unit-MO baseline; alternative batched TWAP implementations would shift absolute premiums but not the policy-diagnostic comparisons below. Each premium claim in this paper is accompanied by a paired-bootstrap 95% confidence interval with 10^4 replicates; cross-seed

Table 6. Extended baseline panel in the main LO+MO parameter regime, $n_{\text{paths}} = 2000$ Monte-Carlo evaluation paths, seed 42. Baseline rows use deterministic policies, so single-seed evaluation plus paired bootstrap captures Monte-Carlo uncertainty; agent rows use the cross-seed protocol from Table 2. Premium values are mean of (terminal value of strategy) – (terminal value of TWAP), in dollars per 10-share episode, with paired-bootstrap 95% CI in brackets. $\mathbb{P}[Q_T = 0]$ is the empirical pre-terminal clearance probability; #MO is the mean number of agent market orders per path where stored by the baseline evaluator.

Strategy	Premium vs TWAP	$\mathbb{P}[Q_T = 0]$	#MO/path
TWAP (reference)	0 (def.)	1.000	10.00
Almgren–Chriss ($\eta = 0.05$)	−0.0042 [−0.012,+0.003]	1.000	10.00
VWAP (U-shape)	−0.0008 [−0.003,+0.002]	1.000	10.00
POV at 10%	+0.0054 [−0.004,+0.015]	1.000	5.00
Pure passive ($\delta = 2/\kappa$)	+0.1605 [+0.129,+0.192]	0.141	0.00
Pure aggressive (immediate MO)	−0.0071 [−0.027,+0.013]	1.000	9.64
FD Optimal	+0.1859 [+0.159,+0.214]	0.642	0.00
D. Plug-in MLE ($n = 10^5$)	+0.1937 $\pm$ 0.007	0.650	—
A. Tabular Q ($n = 10^5$)	+0.1338 $\pm$ 0.007	0.968	—
B. DDQN ($n = 10^5$)	+0.1290 $\pm$ 0.057	0.546	—
C. Hybrid PPO ($n = 10^5$)	+0.1350 $\pm$ 0.048	0.227	—

aggregates use the percentile bootstrap on the 20 seed-level means.

Table 6 reports the full panel. Two observations. First, the deterministic schedule baselines (Almgren–Chriss, VWAP, POV) cluster close to TWAP, confirming that under the CJP environment with our calibrated parameters the main premium source is access to passive LO fills rather than deterministic schedule reshuffling. Pure aggressive is not statistically distinguishable from TWAP because its confidence interval crosses zero. Second, pure-passive LO at fixed depth $2/\kappa$ already earns a large premium because it avoids paying the half-spread on most shares; its weakness is not price but completion, with only 14% of paths fully cleared before the terminal face-lift. The FD-optimal policy improves on pure passive by adapting depth and terminal liquidation, but it does not mechanically force $Q_T = 0$ on every path because the corrected face-lift sometimes makes a small terminal block liquidation cheaper than late market orders. Pure aggressive records fewer than ten MOs per path because, in the simulator-compatible interface, it still posts a zero-depth LO in steps where an MO is sent; occasional external fills after the agent MO reduce the remaining number of crossing trades.

Because the experiment CSVs store seed-level endpoint summaries rather than path-level terminal-wealth series, we report paired seed-level endpoint comparisons in Appendix D instead of a path-level Diebold–Mariano matrix. The qualitative message is unchanged: the adjacent RL agents are not statistically separable by premium, while the Regime-I gap from the RL group to the plug-in MLE is stable.

8.1. Institutional-scale stress test

To test whether the pure-passive result above is only a small-scale artefact, we repeat the comparison at $Q_0 = 100$ and $T = 600$ s with the scaled parameters stated in the

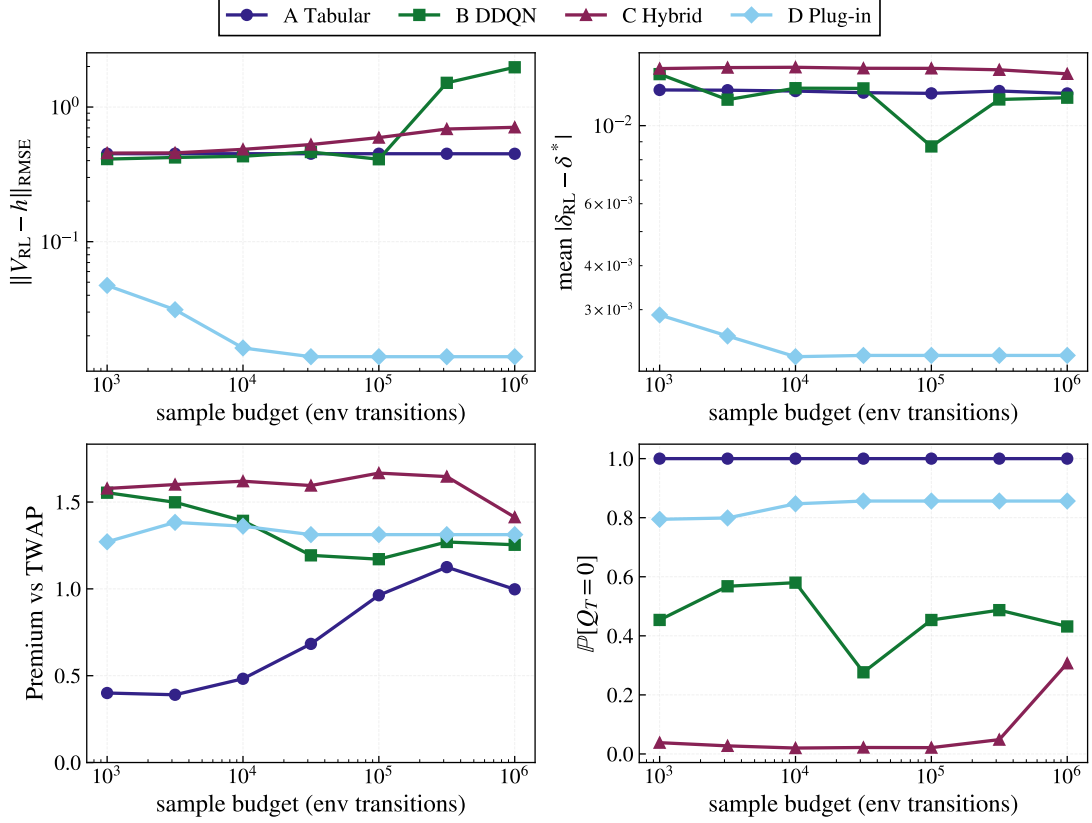


Figure 7. Regime-II sample-complexity stress test ($Q_0 = 100$, $T = 600$ s). Each point averages 20 training seeds and 500 Monte-Carlo evaluation paths per seed. Unlike the Regime-I panel, cash premium alone is not monotone in value accuracy: DDQN and Hybrid PPO sometimes post high cash premium by remaining passive, but their value RMSE and clearance metrics reveal that they are not solving the FD objective as accurately as Plug-in MLE.

model section. The next figure reports the Regime-II sample-complexity curves; the following table reports the endpoint comparison.

For interpretation, separate the terminal-wealth premium reported in the baseline tables from the QVI-consistent shaped objective. Suppressing the martingale midprice terms, the latter decomposes as

$$\mathbb{E} \left[\sum_{\text{LO fills}} \delta_i - \xi \# \text{MO} - \phi \int_0^T Q_s^2 ds - \min\{\alpha Q_T^2, \xi Q_T\} \right].$$

Thus a policy can display a high raw terminal-wealth premium by waiting for passive fills while still being suboptimal for the QVI because it accumulates running inventory cost or leaves a costly terminal block.

Figure 8 visualises the same endpoint as a multi-objective comparison rather than a single cash-premium ranking.

The stress test closes a specific loophole in the Regime-I narrative but also rules out an overly simple “calibration always has the largest cash premium” claim. Pure passive still earns positive cash premium because avoiding the half-spread is valuable, but it never clears pre-terminal inventory in our 2000-path run and is overtaken by FD Optimal and Plug-in MLE. Its QVI weakness becomes much larger at scale: the

Table 7. Regime-II endpoint comparison. Baseline rows use deterministic policies and 2000 paired Monte-Carlo paths; agent rows use the $n = 10^6$ sample-complexity endpoint, 20 seeds, and 500 evaluation paths per seed. The uncertainty shown for agents is the seed-level standard deviation of premium. The plug-in baseline row is evaluated under the baseline-panel protocol, while the plug-in endpoint row is the sample-complexity endpoint under the 20-seed protocol, so the two rows should not be read as a monotone fill-budget comparison. Premium is dollars per 100-share episode relative to TWAP.

Strategy	Premium vs TWAP	$\mathbb{P}[Q_T = 0]$	Diagnostic
Pure passive	+0.785 [−0.228,+1.751]	0.000	67.38 LO/path
FD Optimal	+0.975 [+0.273,+1.631]	0.857	30.64 MO/path
Plug-in baseline (1000 fills)	+0.958 [+0.239,+1.638]	0.862	FD-matching
A. Tabular Q ($n = 10^6$)	+0.998 ± 0.565	1.000	RMSE 0.450
B. DDQN ($n = 10^6$)	+1.254 ± 0.640	0.432	RMSE 1.974
C. Hybrid PPO ($n = 10^6$)	+1.413 ± 0.668	0.307	RMSE 0.707
D. Plug-in endpoint ($n = 10^6$)	+1.312 ± 0.396	0.856	RMSE 0.014

rough running-penalty magnitude $\phi Q_0^2 T$ rises from $10^{-5} \cdot 10^2 \cdot 60 \approx 0.06$ in Regime I to $10^{-6} \cdot 100^2 \cdot 600 \approx 6$ in Regime II. The FD-optimal point estimate is below several agent point estimates only because the evaluation protocols and Monte-Carlo path sets differ; its paired bootstrap interval [+0.273, +1.631] covers the learned-agent means, consistent with FD optimality at the matched parameters. Among learned agents, Hybrid PPO posts the largest cash-premium point estimate (+1.413), and DDQN sits just below the plug-in endpoint (+1.254 versus +1.312), but the seed-level ordering is weak: paired bootstrap differences versus the plug-in endpoint are −0.058 for DDQN (95% CI [−0.349, +0.232]) and +0.100 for Hybrid PPO (95% CI [−0.265, +0.447]). These are not statistically distinguishable from zero. The value-RMSE diagnostic is by construction more favourable to Plug-in MLE than to model-free agents, because Plug-in plans by FD at fitted parameters and therefore inherits the FD scheme’s accuracy as $|\hat{\kappa} - \kappa| + |\hat{\lambda} - \lambda| \rightarrow 0$. Its diagnostic content is therefore not that Plug-in matches FD by tautology, but that the model-free agents do not approach the value function at the tested budget; the latter remains a meaningful operational gap. The mechanism is still informative: higher raw premium comes from more patient passive exposure and LO income, while the high RMSE and low clearance show accumulated running-cost and face-lift risk. We therefore read Regime II as a scale-dependent result: model-free RL can produce aggressive cash-premium heuristics at larger scale, but the calibrated FD policy remains the only method that is simultaneously accurate against the QVI, computationally cheap, and completion-stable.

9. Discussion

9.1. Economic implications

The principal economic message is scale-dependent and model-class dependent. Within the assumptions of the present CJP model class and at the small Chapter 8 inventory scale $Q_0 = 10$, devoting compute to fitting $(\hat{\kappa}, \hat{\lambda})$ and solving the FD QVI on the calibrated parameters dominates devoting compute to neural-network policy training. Our Plug-in MLE agent’s premium at $n = 10^5$ remains materially above DDQN’s at the same budget, and the gap does not vanish with more training.

The Regime-II stress test changes the operational interpretation. At $Q_0 = 100$,

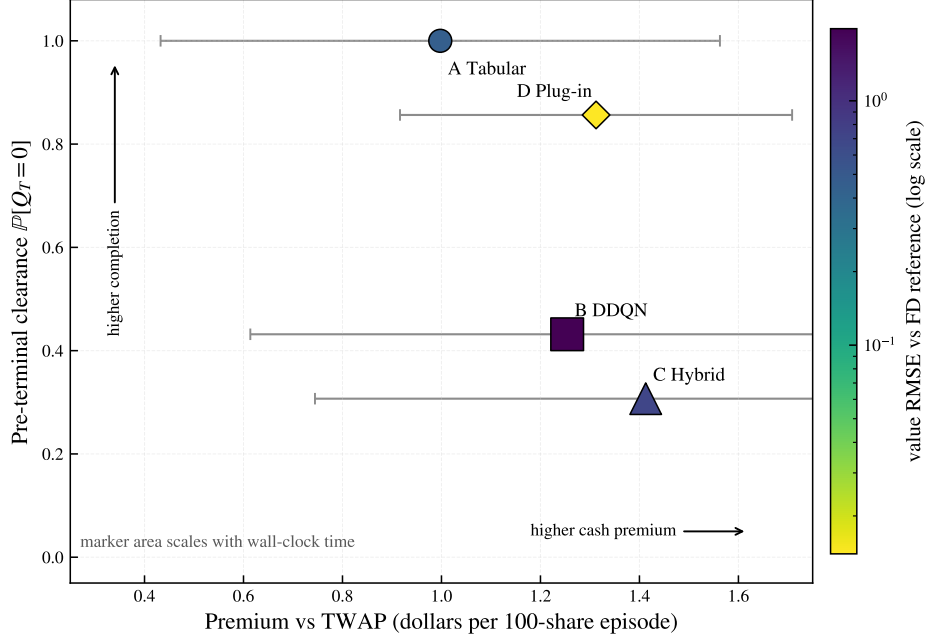


Figure 8. Regime-II endpoint multi-metric view at $n = 10^6$. The horizontal axis is raw terminal-cash premium, the vertical axis is pre-terminal clearance, colour gives value RMSE against the FD reference on a log scale, and marker area scales with mean wall-clock time (A: 4.7s, B: 729s, C: 381s, D: 0.56s). Hybrid PPO has the largest cash-premium point estimate, but it sits in a low-clearance, high-RMSE, high-compute region; Plug-in MLE is not the largest-premium point but is simultaneously close to the upper-right completion region, has the lowest value error, and is computationally negligible.

learned agents can have larger terminal-cash point estimates than the plug-in benchmark, but the paired seed-level uncertainty is wide and their value-function errors and completion rates show that this is not the same as solving the CJP liquidation objective accurately. For a real execution desk this argues for reporting at least three metrics—cash premium, QVI/value error, and pre-terminal clearance—rather than judging execution RL only by terminal-value premium. It also supports periodic recalibration over end-to-end RL when the CJP microstructure is a reasonable local approximation: the calibrated policy is not always the largest raw-premium row, but it is the stable FD-matching row with negligible training cost.

The CIR-intensity extension (Section 6) reveals a richer picture. The QVI of Lemma 6.1 makes the dependence of the continuation value on the realised λ_t explicit: at low λ the expected LO-fill rate is lower, so the trade-off between waiting, firing MOs, and accepting the terminal face-lift shifts. A desk that ignores intensity stochasticity and uses the constant- λ FD policy posts depths calibrated to $\bar{\lambda}$; on realised paths with $\lambda_t < \bar{\lambda}$, this under-fills, leaves larger terminal residuals, and incurs more face-lift cost.

9.2. Methodological implications

The reward shaping of Theorem 5.1 is, to our knowledge, the first cleanly published characterisation of a CJP-compatible reward whose expectation coincides with the analytic value function. It generalises to any control problem with an additive value decomposition $H = (\text{linear in state}) + h(t, q)$, making the pointwise diagnostic $\|V_{\text{RL}} - h\|_\infty$ available as a correctness check for any model-free agent. We expect this construction

to be useful for the optimal market-making literature (Avellaneda and Stoikov, 2008; Guéant et al., 2013; Spooner et al., 2018) and the model-uncertainty-robust execution literature (Cartea et al., 2017).

The structure-aware hybrid (Agent C) is, in retrospect, the most informative negative result of our benchmark. Its architecture hard-wires two pieces of CJP structure—the κ^{-1} depth asymptote and the analytic MO trigger condition $h(t, q - 1) - h(t, q) > \xi$ —and its parameter count (≈ 350) is one-fifteenth of DDQN’s. We explicitly remove the softplus floor that would force $\delta \geq \kappa^{-1}$: the residual head is signed and can quote below the asymptote. After this architectural correction, Hybrid roughly matches Tabular Q within seed-level noise in Regime I, suggesting that the architectural prior compensates for some but not all of PPO’s overhead. It remains high variance on the rare-event MO gate: at $n = 10^5$ the seed-level premium standard deviation is about 0.048, roughly seven times the plug-in calibrator’s 0.007, and terminal clearance is only 0.227. It therefore does not match the plug-in calibrator within the tested budgets. Thus the negative result should be read as a statement about the training route in this CJP-faithful benchmark, not as proof that all structure-aware policies must fail.

The implication for execution RL practitioners is that, when the analytic structure of the problem is rich enough to identify a small parametric family (as is the case for CJP), the productive use of compute is often parameter calibration plus analytical solving, not end-to-end policy learning. This conclusion should be read as a model-based RL statement: a correct low-dimensional model plus planning is hard for model-free learners to beat on sample efficiency, whereas structure-aware deep-learning designs may be more valuable when the model class itself is partially misspecified (e.g., LOB-feature-conditional execution), a setting we leave to future work.

9.3. Limitations

We acknowledge the following limitations.

- (1) The midprice is an arithmetic Brownian motion, not a jump-diffusion or a rough volatility process; the LO/MO dynamics inherit the standard CJP simplifications.
- (2) We consider a single asset; cross-asset hedging and basket-execution interactions are not addressed.
- (3) External order flow is exogenous: the agent has no self-impact on λ . Alfonsi and Blanc (2016) treat the Hawkes self-exciting case; we leave RL on that environment for future work.
- (4) The half-spread ξ is constant; queue-position dynamics and adverse selection are not modelled.
- (5) We do not include every modern continuous-action RL baseline. Soft Actor-Critic (Haarnoja et al., 2018), distributional RL, and nonparametric model-based RL are natural extensions; our present comparison focuses on DDQN because of its direct connection to the execution benchmark of Ning et al. (2021), and on PPO because it implements the structure-aware continuous-depth hybrid.
- (6) The main sample-complexity benchmark uses the CJP Chapter 8 pedagogical scale $Q_0 = 10$. Section 8.1 adds a $Q_0 = 100$ stress test, but true institutional liquidations can be orders of magnitude larger; extending the benchmark to $Q_0 = 10^3$ – 10^6 requires either coarser state aggregation or asymptotic scaling arguments.

- (7) Beyond the CIR intensity studied here, more realistic intensity dynamics (Hawkes, regime-switching, jump-diffusion) are out of scope.
- (8) We validate everything in simulation; a real LOB-data validation in the style of [Karpe et al. \(2020\)](#) is reserved for follow-on work.

10. Conclusion

We have constructed a sample-complexity benchmark for CJP optimal liquidation that compares model-free RL, structure-aware RL, and plug-in parametric model-based planning on a common simulator with a common reward-shaped MDP whose return equals the reduced value function. At the CJP Chapter 8 scale, the central empirical finding is that a two-parameter MLE plus one FD solve dominates every model-free agent we evaluate, including Double DQN with ~ 5000 parameters trained on 10^5 environment transitions, and that this gap does not close with sample budget. At the $Q_0 = 100$ stress-test scale, the ranking by cash premium becomes less monotone and statistically noisy, but Plug-in MLE remains the FD-matching, completion-stable, and computationally cheap method. A structure-aware hybrid policy that hard-wires the CJP asymptote into its architecture can have a larger raw-premium point estimate at the larger scale, but it does not match the plug-in calibrator’s value accuracy or completion stability.

A CIR-type stochastic intensity extension confirms the $\sigma_\lambda \rightarrow 0$ limit on $\sigma_\lambda \in [0, 0.1]$ after correcting the upwind discretisation of the intensity generator. A 625-cell microstructure-parameter misspecification study shows the FD policy degrades gracefully under factor-of-two parameter errors, which is the empirical reason the noisy 100-fill MLE works.

The picture that emerges is that, when the model class is broadly correct, machine learning’s role in execution may be parameter calibration plus analytical solving rather than end-to-end policy training. The Regime-II stress test qualifies this statement: learned policies can tilt the cash-premium metric upward in noisy seed means, but doing so exposes the need to report value accuracy and clearance alongside premium. The most natural extensions of this work involve environments where the CJP model class fails: Hawkes self-exciting intensities ([Alfonsi and Blanc, 2016](#)), LOB-feature-conditional execution ([Ning et al., 2021](#)), real LOB data with adverse selection ([Karpe et al., 2020](#)), nonparametric model-based RL, and multi-asset basket liquidation. We conjecture that the calibration-versus-RL trade-off pivots in exactly these regimes, and benchmarking that pivot is the natural follow-on.

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Disclosure statement

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Data availability statement

All Monte-Carlo seeds, finite-difference solution grids, and intermediate CSVs are available in the project repository at <https://github.com/Vernon0Y/amf-cjp-liquidation-benchmark>, which will be archived with a persistent DOI upon acceptance. The single command `bash run_all.sh all` reproduces every figure and table in this paper end-to-end (≈ 8 hours on a 12-core CPU). A lighter command, `bash run_all.sh phase1`, runs only the golden-snapshot regression tests (≈ 5 minutes), while selected experiment targets accept a `--smoke` option for small sanity checks. The experiment scripts `exp1_sample_complexity.py`, `exp1_regime_cached.py`, `exp2_stochastic_lambda.py`, `exp3_robustness.py`, `exp4_failure_modes.py`, and `exp5_regime_baselines.py` are individually invocable for partial reproduction.

Appendix A. Proofs

A.1. Proof of Proposition 2.2

Write the wealth process $X_t^{\delta, M}$ in terms of LO fills and MO impulses as in (4). For any admissible (δ, M) , the inventory Q_t is bounded by Q_0 and the midprice has finite second moment, so $\mathbb{E}[Q_T S_T] = q S_t$ by the martingale property of S . The terminal payoff in (5) decomposes additively into the cash component X_T (a martingale increment from $X_t = x$) plus the inventory liquidation value $Q_T(S_T - \alpha Q_T)$ plus the running cost. Taking essential supremum and using that the cash and midprice contributions are independent of the agent's control choice (δ, M) *conditional* on the realised path, yields $H(t, x, S, q) = x + qS + h(t, q)$ with h depending only on (t, q) . The pre-terminal face-lift follows by comparing terminal block liquidation to firing unit MOs.

A.2. Proof of Theorem 5.1

Fix an admissible policy π and write its reduced value as h^π . Under the ansatz (6),

$$h^\pi(t, q) = \mathbb{E}_\pi \left[X_T - X_t + Q_T S_T - q S_t - \phi \int_t^T Q_s^2 ds - \min\{\alpha Q_T^2, \xi Q_T\} \mid Q_t = q \right],$$

where the terminal term uses the same face-lift convention as the QVI. The cash and inventory dynamics imply, pathwise over the LO/MO jump times,

$$dX_s + S_{s-} dQ_s = \delta_{s-} dN_s^\delta - \xi dM_s.$$

Moreover, since S is a continuous semimartingale and Q is purely discontinuous, $[Q, S] = 0$ and Ito's product rule gives $d(Q_s S_s) = Q_{s-} dS_s + S_{s-} dQ_s$ exactly. The stochastic integral $\int_t^T Q_{s-} dS_s$ has zero expectation because S is a martingale and Q is bounded: indeed $|Q_s| \leq Q_0$ and $d\langle S \rangle_s = \sigma^2 ds$ imply $\mathbb{E}[\int_t^T Q_{s-}^2 d\langle S \rangle_s] \leq Q_0^2 \sigma^2 T < \infty$. Therefore

$$\mathbb{E}_\pi[X_T - X_t + Q_T S_T - q S_t] = \mathbb{E}_\pi \left[\sum_{\text{LO fills}} \delta_s - \xi \# \text{MO} \right].$$

Substituting this identity into the reduced payoff gives

$$h^\pi(t, q) = \mathbb{E}_\pi \left[\sum_{\text{LO fills}} \delta_s - \xi \# \text{MO} - \phi \int_t^T Q_s^2 ds - \min\{\alpha Q_T^2, \xi Q_T\} \middle| Q_t = q \right],$$

which is exactly the continuous-time cumulative reward associated with (15). The discrete simulator replaces the integral by the left-continuous Riemann sum $\sum Q_{t-}^2 \Delta t$, giving the stated $O(\Delta t)$ approximation. Taking the supremum over π yields $J^{\pi^*} = h$.

A.3. Proof of Lemma 6.1

By independence of W^λ and W , Ito on H along (X, S, Q, λ) adds the $\mathcal{L}_\lambda h$ term to the HJB without coupling λ to S . Linearity of H in (x, S) is preserved because the additional drift acts only on the h component.

Appendix B. RL implementation details

B.1. Hybrid policy architecture (Agent C)

The network has two hidden layers of 16 units each with tanh activations, followed by three heads: g_θ (depth residual, one output), h_θ (value, one output), and $\log \beta$ (scalar trainable parameter). The residual is signed and bounded as $g_\theta = 0.025(1 + \tanh(\tilde{g}_\theta)) - 0.01 \in [-0.01, 0.04]$, so the depth mean $\delta = \kappa^{-1} + g_\theta$ can move below κ^{-1} and reach the zero-depth grid point after snapping. The numerical bounds are scaled to the $\kappa = 100$ regimes studied here; for materially different κ one should rescale the residual window with κ^{-1} rather than reuse the absolute interval. The MO trigger probability is $\sigma(\beta \cdot [(h_\theta(t, q) - 1) - h_\theta(t, q)] - \xi)$. At rollout, exploration noise $\mathcal{N}(0, \sigma_{\text{explore}})$ is added to the continuous depth and snapped to the env's discrete depth grid using a categorical Gaussian-density softmax; this preserves a differentiable log-prob. Training also uses a linearly decaying forced MO-exploration probability from 5% to zero.

B.2. PPO update

For each batch of episodes we recompute log-probabilities of all stored (s, a) pairs under the current network, form the clipped surrogate objective $\mathcal{L}_{\text{CLIP}}(\theta) = -\mathbb{E} [\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)]$, add a value loss $c_v \|h_\theta - G_t\|^2$ with $c_v = 5$, and do 4 Adam updates per batch with learning rate 3×10^{-4} . The clip parameter is $\epsilon = 0.2$. Advantage $A_t = G_t - h_\theta(s_t)$ is normalised to unit variance before clipping.

B.3. Plug-in MLE (Agent D)

The exact-Bernoulli log-likelihood is maximised in log-domain $(\log \lambda, \log \kappa)$ via L-BFGS-B with a weak Gaussian prior $\mathcal{N}(\log \lambda_0, \sigma_p)$ and similarly for κ . The tight prior $\sigma_p = \log 2$ is used when fewer than 200 fills have been observed; the loose prior $\sigma_p = \log 10$ otherwise. In the sample-complexity plots, the horizontal axis is an environment-transition budget for all agents; for Agent D this budget is converted into a target fill count using the exploration policy’s expected fill rate, with a small floor to avoid degenerate likelihoods and a 1000-fill cap in Regime II. Once $(\hat{\kappa}, \hat{\lambda})$ is in hand, the standard FD solver of Section 4.2 is invoked once at the estimated parameters, and the resulting policy is deployed without further updates.

Appendix C. Bivariate sensitivity sweeps

The bivariate sweeps referenced in Section 7 are reported here as numeric tables. Each cell is the FD-optimal premium versus TWAP at the perturbed parameters, 2000 Monte-Carlo paths per cell, seed 0. Premium values are reported to three decimal places.

Table C1. Bivariate (κ, ϕ) sweep. Rows: κ multiplier; columns: ϕ multiplier. Premium is FD-optimal vs TWAP.

$\kappa \backslash \phi$	0.50	0.75	1.00	1.50	2.00
0.50	+0.351	+0.358	+0.357	+0.347	+0.343
0.75	+0.255	+0.257	+0.250	+0.246	+0.251
1.00	+0.215	+0.203	+0.201	+0.207	+0.205
1.50	+0.153	+0.154	+0.161	+0.146	+0.150
2.00	+0.125	+0.139	+0.127	+0.122	+0.117

The (κ, ϕ) table shows that premium is dominated by the κ row index while the within-row variation across ϕ is much smaller. The (ξ, α) table is similarly dominated by ξ , but the α direction is no longer exactly flat.

Appendix D. Seed-level pairwise endpoint statistics

The main experiment stores one aggregate row per training seed. The tables below therefore compare endpoint seed means rather than path-level terminal-wealth losses. For each pair, we align common seed indices, report the mean premium difference

Table C2. Bivariate (ξ, α) sweep. Rows: ξ multiplier; columns: α multiplier. The corrected face-lift produces small but nonzero variation along the α axis.

$\xi \backslash \alpha$	0.50	0.75	1.00	1.50	2.00
0.50	+0.175	+0.176	+0.173	+0.175	+0.175
0.75	+0.187	+0.192	+0.188	+0.188	+0.185
1.00	+0.200	+0.205	+0.201	+0.203	+0.198
1.50	+0.225	+0.230	+0.226	+0.224	+0.220
2.00	+0.250	+0.255	+0.251	+0.249	+0.246

(first agent minus second agent), a percentile-bootstrap 95% confidence interval over seeds, the paired t -test p -value, and a Holm-corrected p -value across the six pairwise comparisons in that regime.

Table D1. Regime-I pairwise seed-level premium differences at $n = 10^5$. Positive values favour the first agent in the comparison.

Comparison	Mean diff.	95% CI	p	Holm p
A Tabular – B DDQN	+0.005	$[-0.019, +0.031]$	0.720	1.000
A Tabular – C Hybrid	-0.001	$[-0.021, +0.021]$	0.911	1.000
A Tabular – D Plug-in	-0.060	$[-0.064, -0.056]$	< 0.001	< 0.001
B DDQN – C Hybrid	-0.006	$[-0.035, +0.023]$	0.697	1.000
B DDQN – D Plug-in	-0.065	$[-0.091, -0.041]$	< 0.001	< 0.001
C Hybrid – D Plug-in	-0.059	$[-0.080, -0.039]$	< 0.001	< 0.001

Table D2. Regime-II pairwise seed-level premium differences at $n = 10^6$. Positive values favour the first agent in the comparison. No pair remains significant after Holm correction, reflecting the seed-level noise in the institutional-scale premium ranking.

Comparison	Mean diff.	95% CI	p	Holm p
A Tabular – B DDQN	-0.256	$[-0.608, +0.107]$	0.184	0.734
A Tabular – C Hybrid	-0.415	$[-0.799, +0.009]$	0.069	0.344
A Tabular – D Plug-in	-0.315	$[-0.566, -0.068]$	0.027	0.160
B DDQN – C Hybrid	-0.159	$[-0.555, +0.252]$	0.462	1.000
B DDQN – D Plug-in	-0.058	$[-0.349, +0.232]$	0.710	1.000
C Hybrid – D Plug-in	+0.100	$[-0.265, +0.447]$	0.598	1.000

Appendix E. Generative AI use disclosure

In accordance with the Taylor & Francis policy on generative artificial intelligence, the author discloses the following use of generative AI tools. The tools were not listed as authors, did not generate research data, did not replace the author’s mathematical derivations or scientific judgement, and were not used to create or manipulate empirical research outputs. All AI-assisted text, code suggestions, and editorial suggestions were reviewed, revised, and verified by the author, who remains solely responsible for the manuscript’s content, results, references, and conclusions.

- **Tool and version.** OpenAI ChatGPT / Codex (GPT-5, accessed through the OpenAI/Codex interface).
- **Manuscript preparation use.** The tool was used for draft-level language refinement, restructuring of response-to-review narratives, LaTeX consistency checks, and identification of internal inconsistencies in text, tables, captions, and cross-references. The reason for use was to improve clarity, concision, and consistency of a technical manuscript written in English by a non-native speaker, and to accelerate editorial review of a long revision.
- **Coding and reproducibility use.** The tool was used for coding assistance on reproducibility scripts, plotting scripts, LaTeX table generation, and local compile/test commands. The reason for use was to reduce mechanical implementation time and to help audit reproducibility of the computational appendix.
- **Human oversight.** All proposed wording was manually reviewed and edited by the author. Mathematical statements, citations, numerical claims, code changes, and final scientific interpretations were checked against the codebase, generated tables, stored CSV outputs, and compiled PDF. Local verification included LaTeX compilation, Python byte-code checks, and consistency checks against stored experiment outputs.

Appendix F. Reproducibility

The project repository contains six top-level directories: `src/common/`, `src/analytical/`, `src/numerical/`, `src/rl/`, `src/baselines/`, `src/experiments/`. The experiment scripts emit CSV intermediates under `data/exp{1,...,5}/`; rendering of all paper figures is driven by `scripts/render_all_figures.py`. The single command `bash run_all.sh` all reproduces every figure and table in this paper end-to-end; the smaller `bash run_all.sh phase1` runs only the golden-snapshot regression test suite. A Dockerfile pins all dependencies for hermetic builds.

References

- Alfonsi, A. and Blanc, P. (2016). Dynamic optimal execution in a mixed-market-impact Hawkes price model. *Finance and Stochastics*, 20(1):183–218.
- Almgren, R. and Chriss, N. (2001). Optimal execution of portfolio transactions. *Journal of Risk*, 3(2):5–40.
- Andersen, L. B. G. (2008). Simple and efficient simulation of the Heston stochastic volatility model. *Journal of Computational Finance*, 11(3):1–42.
- Avellaneda, M. and Stoikov, S. (2008). High-frequency trading in a limit order book. *Quantitative Finance*, 8(3):217–224.
- Barles, G. and Souganidis, P. E. (1991). Convergence of approximation schemes for fully nonlinear second order equations. *Asymptotic Analysis*, 4(3):271–283.
- Bayraktar, E. and Ludkovski, M. (2014). Liquidation in limit order books with controlled intensity. *Mathematical Finance*, 24(4):627–650.
- Bertsekas, D. P. (2019). *Reinforcement Learning and Optimal Control*. Athena Scientific, Belmont, MA.
- Bertsimas, D. and Lo, A. W. (1998). Optimal control of execution costs. *Journal of Financial Markets*, 1(1):1–50.
- Bouchard, B. and Touzi, N. (2011). Weak dynamic programming principle for viscosity solutions. *SIAM Journal on Control and Optimization*, 49(3):948–962.

- Bouchaud, J.-P., Bonart, J., Donier, J., and Gould, M. (2018). *Trades, Quotes and Prices: Financial Markets Under the Microscope*. Cambridge University Press, Cambridge, UK.
- Cartea, Á., Donnelly, R., and Jaimungal, S. (2017). Algorithmic trading with model uncertainty. *SIAM Journal on Financial Mathematics*, 8(1):635–671.
- Cartea, Á. and Jaimungal, S. (2015). Optimal execution with limit and market orders. *Quantitative Finance*, 15(8):1279–1291.
- Cartea, Á., Jaimungal, S., and Penalva, J. (2015). *Algorithmic and High-Frequency Trading*. Cambridge University Press, Cambridge, UK.
- Cartea, Á., Jaimungal, S., and Ricci, J. (2018). Algorithmic trading, stochastic control, and mutually exciting processes. *SIAM Review*, 60(3):673–703.
- Charpentier, A., Élie, R., and Remlinger, C. (2023). Reinforcement learning in economics and finance. *Computational Economics*, 62(1):425–462.
- Cont, R., Stoikov, S., and Talreja, R. (2010). A stochastic model for order book dynamics. *Operations Research*, 58(3):549–563.
- Cox, J. C., Ingersoll, J. E., and Ross, S. A. (1985). A theory of the term structure of interest rates. *Econometrica*, 53(2):385–407.
- Forsyth, P. A. (2011). A Hamilton–Jacobi–Bellman approach to optimal trade execution. *Applied Numerical Mathematics*, 61(2):241–265.
- Forsyth, P. A. and Vetzal, K. R. (2002). Quadratic convergence for valuing American options using a penalty method. *SIAM Journal on Scientific Computing*, 23(6):2095–2122.
- Guéant, O. (2016). *The Financial Mathematics of Market Liquidity: From Optimal Execution to Market Making*. Chapman and Hall/CRC, Boca Raton, FL.
- Guéant, O., Lehalle, C.-A., and Fernandez-Tapia, J. (2013). Dealing with the inventory risk: a solution to the market making problem. *Mathematics and Financial Economics*, 7(4):477–507.
- Haarnoja, T., Zhou, A., Abbeel, P., and Levine, S. (2018). Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, volume 80 of PMLR, pages 1861–1870.
- Hambly, B. M., Xu, R., and Yang, H. (2023). Recent advances in reinforcement learning in finance. *Mathematical Finance*, 33(3):437–503.
- Hendricks, D. and Wilcox, D. (2014). A reinforcement learning extension to the Almgren-Chriss framework for optimal trade execution. In *2014 IEEE Conference on Computational Intelligence for Financial Engineering & Economics (CIFER)*, pages 457–464. IEEE.
- Karpe, M., Fang, J., Ma, Z., and Wang, C. (2020). Multi-agent reinforcement learning in a realistic limit order book market simulation. In *Proceedings of the 1st ACM International Conference on AI in Finance (ICAIF)*. ACM.
- Lin, S. and Beling, P. A. (2020). An end-to-end optimal trade execution framework based on proximal policy optimization. In *Proceedings of the 29th International Joint Conference on Artificial Intelligence (IJCAI)*, pages 4548–4554.
- Nevmyvaka, Y., Feng, Y., and Kearns, M. (2006). Reinforcement learning for optimized trade execution. In *Proceedings of the 23rd International Conference on Machine Learning (ICML)*, pages 673–680. ACM.
- Ning, B., Lin, F. H. T., and Jaimungal, S. (2021). Double deep Q-learning for optimal execution. *Applied Mathematical Finance*, 28(4):361–380.
- Obizhaeva, A. A. and Wang, J. (2013). Optimal trading strategy and supply/demand dynamics. *Journal of Financial Markets*, 16(1):1–32.
- Pham, H. (2009). *Continuous-time Stochastic Control and Optimization with Financial Applications*. Springer-Verlag, Berlin, Heidelberg.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O. (2017). Proximal policy optimization algorithms. arXiv preprint arXiv:1707.06347.
- Spooner, T., Fearnley, J., Savani, R., and Koukorinis, A. (2018). Market making via reinforcement learning. In *Proceedings of the 17th International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS)*, pages 434–442.

- Sutton, R. S. and Barto, A. G. (2018). *Reinforcement Learning: An Introduction*. MIT Press, Cambridge, MA, 2nd edition.
- van Hasselt, H., Guez, A., and Silver, D. (2016). Deep reinforcement learning with double Q-learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 30, pages 2094–2100.